A. Course Handout (Version 1.0)

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Cloud Computing-II	Session	2025-2026
Course Code	23CS022	Semester/Batch	6th/2023
L-T-P (Per Week)	2-0-2	Course Credits	03
Pre-requisite	Basic concepts of Programming, Storage, Databases	NHEQF Level	5.5
Course Coordinator	Dr. Priyanka Gupta	SDG Number	4,8,9

CLO01	Understand the fundamental concepts and market perspective of cloud computing.
CLO02	Gain familiarity with various AWS tools and services.
CLO03	Perform hands-on practice on the AWS platform.
CLO04	Effectively manage, upscale, and downscale a business using AWS.
CLO05	Design and deploy secure and scalable cloud-based solutions.

1. Objectives of the Course

This course aims to introduce students to cloud computing from fundamental concepts to hands-on practical implementation. By the end of the course, learners will be able to:

I. Understand major cloud platforms and distinguish between public, private, and hybrid cloud models.
II. Learn how to get started with cloud platforms and gain foundational knowledge of widely used cloud infrastructure, particularly Amazon Web Services (AWS).
III. Develop practical skills on the AWS platform to design, customize, and manage cloud-based solutions for diverse business requirements.
IV. Monitor, manage, and optimize cloud resources efficiently using AWS management tools.
V. Apply AWS tagging strategies for effective cost management, security control, and automation.

2. Course Learning Outcomes

After completion of the course, student should be able to:

	Course Learning Outcome	*Pos	**CL	***KC	Sessions
CLO01	Understand the fundamental concepts and market perspective of cloud computing.	PO1, PO3, PO4, PO8, PO12	K2	Factual Conceptual	**16**
CLO02	Gain familiarity with various AWS tools and services.	PO1, PO3, PO4, PO5, PO8, PO12	K3	Conceptual Procedural	**16**

CLO03	Perform hands-on practice on the AWS platform.	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO10, PO11, PO12	K3	Conceptual Procedural	20
CLO04	Effectively manage, upscale, and downscale a business using AWS.	PO1, PO2, PO3, PO4, PO5, PO7, PO8, PO10, PO11, PO12	K4	Conceptual Procedural	**18**
CLO05	Design and deploy secure and scalable cloud-based solutions.	PO1, PO2, PO3, PO4 ,PO6, PO8, PO9, PO11, PO12	K3	Conceptual Procedural	**20**
Total Contact Hours					**90**

Revised Bloom's Taxonomy Terminology

* PO's available at (shorturl.at/cryzF)

**Cognitive Level =CL

***Knowledge Categories = KC

Course Learning Outcomes	**PO 1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CLO01	H	H	H	M					L		L	H
CLO02	H	H	H	M					L		L	H
CLO03	H	H	H	H					M		M	H
CLO04	H	H	H	H	M				H		M	H
CLO05	H	H	H	H	M				H		M	H

H=High, M=Medium, L=Low

3. ERISE Grid Mapping

Feature Enablement	**Level (1-5, 5 being highest)**
Entrepreneurship	4
Research	3
Innovation	5
Skills	5
Employability	5

4. Recommended Books (Reference Books/Textbooks)

B01: Architecting the Cloud: Design Decisions for Cloud Computing Service Models by Michael J. Kavis, 2014.

BO2: AWS The Ultimate Guide to Amazon Web Services Step-by-step Guide from Beginners to Advanced! by Stephen Springwell, Amplitude Ltd., Feb-2021.

BO3: Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud,

First Edition by Pearson (Mark Wilkins), Nov-2019.

B04: Thomas Erl, Zaigham Mahmood, and Ricardo Puttini, *Cloud Computing: Concepts, Technology & Architecture*, Pearson Education.

B05: Michael Miller, Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online, Pearson.

E-Resources:

AWS - https://aws.amazon.com/

https://library.chitkara.edu.in/

5. Other readings & relevant websites

S.N.	Link of Journals, Magazines, Websites, and Research Papers
1	https://computingforgeeks.com/how-to-learn-cloud-computing-from-beginner-to-expert/#:~:text=Learn%20Cloud%20Computing%20from%20Scratch%201%20Step%20One%3A,Six%3A%20Get%20Deep%20in%20your%20area%20of%20expertise
2	https://aws.amazon.com/free/database/?trk=ce743268-ef4c-43aa-9909-188edfb237f9&sc_channel=ps&s_kwcid=AL!4422!10!71331043936464!71331563897761&s_kwcid=AL!4422!10!71331043936464!71331563897761&ef_id=3129036f1e2f156b980c608019841b12:G:s
3	https://www.simplilearn.com/cloud-solutions-architect-masters-program-training?utm_source=bing&utm_medium=cpc&utm_term=simplilearn%20AWS%20training&utm_content=369201188-1329310345774847-&utm_device=c&utm_campaign=B-Search-Brand-Broad-IN-AllDevice-adgroup-Brand-Cat-AWS&msclkid=27c934140d0110e2801257fb0aaa501c

6. Recommended Tools and Platforms

AWS - https://aws.amazon.com/

7. Course Plan

Lecture Number	Topics	Recommended Books
1-4	**Cloud Architecture:** Layers and components of cloud services, Deployment Models: Public, Private, Hybrid, Community with comparison, Service Models: IaaS, PaaS, SaaS and their differences, Cloud Data Lifecycle.	B01–Chapter–1,2; B04–Chapter–1,2; B05–Chapter–1
5-9	**Virtualization:** Introduction to virtualization techniques, Characteristics of virtualization, Pros and Cons of virtualization technology, Hypervisors and Types of Hypervisors, Multitenancy, Elasticity and Scalability, VirtualBox.	B01–Chapter–3,4; B04–Chapter–3,4; B05–Chapter–2
10-13	**AWS Learner Lab Environment & Console:** Overview of AWS Academy Learner Lab, Lab session duration, budgets and limitations, Starting, stopping and resetting lab sessions, AWS Management Console overview, Regions and Availability Zones.	B01–Chapter–7,8; B03–Chapter–3,4; B04–Chapter–6; B05–Chapter–4
14-20	**Identity and Access Management (IAM) :** Introduction to IAM concepts, Users, groups, roles and policies (conceptual overview), Preconfigured IAM roles in Learner Lab, Root account restrictions, Shared Responsibility Model.	B02–Chapter–1,2,3; B03–Chapter–5,6; B04–Chapter–7; B05–Chapter–5

21-26	**Compute Services:** Introduction to Amazon EC2, Instance types, Launching and managing EC2 instances, Instance lifecycle management, Amazon Machine Images (AMIs), Auto Scaling, Elastic IP, Elastic Load Balancer and its Types, AWS CloudShell.	B02–Chapter–4,5,6; B03–Chapter–7,8; B04–Chapter–8; B05–Chapter–6
27-33	**Storage Services:** Amazon S3 overview, Buckets and objects, S3 storage classes, Basic S3 permissions, Amazon EBS volumes, Attaching and managing EBS with EC2, Amazon EFS, Creating and mounting EFS with EC2, Amazon Glacier Storage.	B02–Chapter–7,8,9; B03–Chapter–9,10; B04–Chapter–9; B05–Chapter–7
34-40	**Containerization:** Introduction to containerization, Docker concepts and architecture, Containers vs Virtual Machines, Kubernetes, Cloud security principles, Data protection strategies, Backup and recovery concepts, Cost awareness in AWS.	B02–Chapter–10,11,12;B03–Chapter–11,12; B04–Chapter–10; B05–Chapter–8
41-45	**Networking Services:** Introduction to AWS Networking, Virtual Private Cloud (VPC), Subnets and CIDR blocks, Route Tables and Internet Gateway, Security Groups, Network Access Control Lists (NACLs), Route 53 (DNS System), AWS Direct Connect, Transit Gateway, CloudFront.	B01–Chapter–11,12; B02–Chapter–15,16; B03–Chapter–15,16; B05–Chapter–9
	ST-1 (Syllabus covered from Lecture 1 to 45)	
46-53	**AWS Security & Application Deployment:** AWS Security: Introduction to AWS Security, AWS Key Management Service (KMS), Securing data at rest and in motion, Compliance issues, Privacy and security.	B01–Chapter–13,14; B02–Chapter–17,18; B03–Chapter–17; B04–Chapter–12
54-58	**Application Deployment:** AWS Elastic Beanstalk overview, Application deployment workflow, Environment creation and management, Application versioning basics.	B01–Chapter–15,16; B02–Chapter–19,20; B03–Chapter–18; B05–Chapter–10
59-63	**Messaging and Integration Services:** Amazon Simple Notification Service (SNS), Amazon Simple Queue Service (SQS), Event-driven architecture concepts, Message-based communication, AWS CloudTrail, AWS CloudWatch and Alarms.	B01–Chapter–17,18; B02–Chapter–21,22; B03–Chapter–19; B04–Chapter–13
64-69	**Introduction to DevOps in Cloud:** Definition and principles of DevOps, Role of DevOps in cloud environments, Key DevOps practices: CI/CD, Automation, Monitoring, Collaboration.	B01–Chapter–19,20; B02–Chapter–23,24; B03–Chapter–20; B05–Chapter–11
70-80	**Database Services:** Overview of AWS managed databases, Amazon RDS (MySQL /PostgreSQL), Database creation and connectivity, Amazon DynamoDB fundamentals, Data persistence across lab sessions, Amazon Redshift, Amazon Aurora	B02–Chapter–25,26,27; B04–Chapter–14,15; B05–Chapter–12

81-85	**Data Analytics Tools in AWS:** Amazon Athena, AWS Glue, Amazon Quick Sight, Amazon EMR, AWS Data Pipeline, Lake Formation, Amazon Kinesis, Amazon Timestream, Amazon SageMaker.	B01–Chapter–7,8; B03–Chapter–3,4; B04–Chapter–6; B05–Chapter–4
86-90	**AWS Developer & Migration Tools:** AWS X-Ray, AWS Cloud9, AWS CodeBuild, AWS CodeCommit, AWS CodePipeline, Introduction to Migration Tools, AWS Database Migration Service (DMS), AWS Server Migration Service (SMS), AWS Migration Hub.	B02–Chapter–1,2,3;B03–Chapter–5,6; B04–Chapter–7; B05–Chapter–5
	ST-2 (Syllabus covered from Lecture 46 to 90)	
END TERM – FULL SYLLABUS		

8. Delivery/Instructional Resources

Lecture No.	Topics	Web References	Audio-Video
1-4	**Cloud Architecture:** Layers and components of cloud services, Deployment Models: Public, Private, Hybrid, Community with comparison, Service Models: IaaS, PaaS, SaaS and their differences, Cloud Data Lifecycle.	• https://aws.amazon.com/what-is-cloud-computing/ • https://aws.amazon.com/types-of-cloud-computing/ • https://aws.amazon.com/what-is/iaas/ • https://aws.amazon.com/what-is/paas/ • https://aws.amazon.com/what-is/saas/ • https://www.ibm.com/topics/cloud-lifecycle	https://www.youtube.com/watch?v=HK_q1lH5x5M https://www.youtube.com/playlist?list=PLEiEAq2VkUULlNtIFhEQHo8gacvme35rz
5-9	**Virtualization:** Introduction to virtualization techniques, Characteristics of virtualization, Pros and Cons of virtualization technology, Hypervisors and Types of Hypervisors, Multitenancy, Elasticity and Scalability, VirtualBox.	https://www.redhat.com/en/topics/virtualization	https://youtu.be/fZ3D6HQrWzs

10-13	**AWS Learner Lab Environment & Console:** Overview of AWS Academy Learner Lab, Lab session duration, budgets and limitations, Starting, stopping and resetting lab sessions, AWS Management Console overview, Regions and Availability Zones.	https://aws.amazon.com/console/ https://docs.aws.amazon.com/general/latest/gr/aws-regions-and-availability-zones.html	https://www.youtube.com/playlist?list=PLdpzxOOAlwvLNOxX0RfndiYSt1Le9azze
14-20	**Identity and Access Management (IAM) :** Introduction to IAM concepts, Users, groups, roles and policies (conceptual overview), Preconfigured IAM roles in Learner Lab, Root account restrictions, Shared Responsibility Model.	https://aws.amazon.com/iam/ https://docs.aws.amazon.com/IAM/latest/UserGuide/introduction.html	https://www.youtube.com/watch?v=1LV_ZuUSQiw
21-26	Compute Services: Introduction to Amazon EC2, Instance types, Launching and managing EC2 instances, Instance lifecycle management, Amazon Machine Images (AMIs), Auto Scaling, Elastic IP, Elastic Load Balancer and its Types, AWS CloudShell.	https://aws.amazon.com/ec2/ https://docs.aws.amazon.com/ec2/ https://aws.amazon.com/autoscaling/ https://aws.amazon.com/elasticloadbalancing/ https://aws.amazon.com/cloudshell/	https://www.youtube.com/watch?v=9JTzOOlkepY&t=2s
27-33	Storage Services : Amazon S3 overview, Buckets and objects, S3 storage classes, Basic S3 permissions, Amazon EBS volumes, Attaching and managing EBS with EC2, Amazon EFS, Creating and mounting EFS with EC2, Amazon Glacier Storage.	https://aws.amazon.com/s3/ https://docs.aws.amazon.com/AmazonS3/latest/userguide/ https://aws.amazon.com/ebs/ https://aws.amazon.com/efs/ https://aws.amazon.com/glacier/	https://www.youtube.com/watch?v=A2N9OIun9dU https://youtu.be/oaZ3R4NCRu8

34-40	Containerization: Introduction to containerization, Docker concepts and architecture, Containers vs Virtual Machines, Kubernetes, Cloud security principles, Data protection strategies, Backup and recovery concepts, Cost awareness in AWS.	https://www.docker.com/resources/what-container/ https://kubernetes.io/docs/concepts/ https://aws.amazon.com/security/ https://aws.amazon.com/backup/ https://aws.amazon.com/aws-cost-management/	https://www.youtube.com/watch?v=UT88Ojx-TLk
41-45	Networking Services: Introduction to AWS Networking, Virtual Private Cloud (VPC), Subnets and CIDR blocks, Route Tables and Internet Gateway, Security Groups, Network Access Control Lists (NACLs), Route 53 (DNS System), AWS Direct Connect, Transit Gateway, CloudFront.	https://aws.amazon.com/vpc/ https://docs.aws.amazon.com/vpc/latest/userguide/ https://aws.amazon.com/route53/ https://aws.amazon.com/directconnect/ https://aws.amazon.com/cloudfront/ https://aws.amazon.com/transit-gateway/	https://www.youtube.com/playlist?list=PLEiEAq2VkUUKR5bCWqRp2v_-8hhOEST0H https://youtu.be/b1b6JTYnbjU
46-53	AWS Security & Application Deployment : AWS Security: Introduction to AWS Security, AWS Key Management Service (KMS), Securing data at rest and in motion, Compliance issues, Privacy and security.	https://aws.amazon.com/kms/ https://aws.amazon.com/security/ https://aws.amazon.com/compliance/	https://www.youtube.com/watch?v=TS0UhSfzy_4
54-58	Application Deployment: AWS Elastic Beanstalk overview, Application deployment workflow, Environment creation and management, Application versioning basics.	https://aws.amazon.com/elasticbeanstalk/ https://docs.aws.amazon.com/elasticbeanstalk/	https://www.youtube.com/watch?v=yjHmzo91Rek

59-63	Messaging and Integration Services: Amazon Simple Notification Service (SNS), Amazon Simple Queue Service (SQS), Event-driven architecture concepts, Message-based communication, AWS CloudTrail, AWS CloudWatch and Alarms.	https://aws.amazon.com/sns/ https://aws.amazon.com/sqs/ https://aws.amazon.com/cloudwatch/ https://aws.amazon.com/cloudtrail/ https://aws.amazon.com/eventbridge/	https://www.youtube.com/watch?v=Xn2C-nPcSio
64-69	Introduction to DevOps in Cloud: Definition and principles of DevOps, Role of DevOps in cloud environments, Key DevOps practices: CI/CD, Automation, Monitoring, Collaboration.	https://aws.amazon.com/devops/ https://aws.amazon.com/devops/what-is-devops/ https://docs.aws.amazon.com/codepipeline/ https://aws.amazon.com/cicd/	https://www.youtube.com/watch?v=JRmV_mKnnyU
70-80	Database Services: Overview of AWS managed databases, Amazon RDS (MySQL / PostgreSQL), Database creation and connectivity, Amazon DynamoDB fundamentals, Data persistence across lab sessions, Amazon Redshift, Amazon Aurora	https://aws.amazon.com/rds/ https://aws.amazon.com/dynamodb/ https://aws.amazon.com/redshift/ https://aws.amazon.com/rds/aurora/ https://docs.aws.amazon.com/rds/	https://www.youtube.com/watch?v=ew3mwz-tOeg
81-85	Data Analytics Tools in AWS: Amazon Athena, AWS Glue, Amazon QuickSight, Amazon EMR, AWS Data Pipeline, Lake Formation, Amazon Kinesis, Amazon Timestream, Amazon SageMaker.	https://aws.amazon.com/athena/ https://aws.amazon.com/glue/ https://aws.amazon.com/quicksight/ https://aws.amazon.com/emr/ https://aws.amazon.com/kinesis/ https://aws.amazon.com/sagemaker/	https://www.youtube.com/watch?v=z0Owwd3u6Dw https://www.youtube.com/results?search_query=Amazon+QuickSight+tutorial

86-90	AWS Developer & Migration Tools : AWS X-Ray, AWS Cloud9, AWS CodeBuild, AWS CodeCommit, AWS CodePipeline, Introduction to Migration Tools, AWS Database Migration Service (DMS), AWS Server Migration Service (SMS), AWS Migration Hub.	https://aws.amazon.com/cloud9/ https://aws.amazon.com/codebuild/ https://aws.amazon.com/codecommit/ https://aws.amazon.com/codepipeline/ https://aws.amazon.com/dms/ https://aws.amazon.com/migration-hub/ https://aws.amazon.com/xray/	https://www.youtube.com/watch?v=PY8UuYb-Wfc&t=5s

9. Lab Plan

Lab Number	Practical	Learning Resources
1-8	Installation, Configuration, and Management of Virtual Machines Using VirtualBox	https://www.virtualbox.org/manual
9-20	Creation, Secure Launch, Network Configuration, and Custom AMI Generation for Amazon EC2 Instances Using Key Pairs and Security Groups.	https://docs.aws.amazon.com/ec2/
21-26	Installation, Configuration, and Deployment of Web Applications on EC2 Using Apache/Nginx Web Server.	https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-key-pairs.html
27-35	Implementation and Management of Cloud Storage Services and Static Website Hosting Using Amazon S3, EBS, and EFS.	https://docs.aws.amazon.com/s3/ https://docs.aws.amazon.com/ebs/ https://docs.aws.amazon.com/efs/
36-45	Installation, Configuration, and Deployment of Containerized Applications Using Docker on Amazon EC2.	https://docs.docker.com/
Lab Practical (Syllabus covered from 1 to 45)		
46-50	Design and Implementation of Auto Scaling, Load Balancing, and Elastic IP for High Availability and Scalability.	https://docs.aws.amazon.com/elasticloadbalancing/ https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/elastic-ip-addresses-eip.html

51-58	Deployment, Monitoring, and Auto-Scaling of Web Applications Using AWS Elastic Beanstalk Platform	https://docs.aws.amazon.com/autoscaling/ https://docs.aws.amazon.com/elasticbeanstalk/
59-63	Monitoring, Automation, and Messaging Using Amazon CloudWatch, AWS Cloud Shell, SNS, and SQS.	https://docs.aws.amazon.com/cloudwatch/
64-72	Design, Configuration, and Management of AWS Networking Services Including VPC, Subnets, Route Tables, Security, DNS and CloudFront	https://docs.aws.amazon.com/vpc/ https://docs.aws.amazon.com/cloudfront/
73-90	Exploring AWS Managed Databases Using Amazon RDS and DynamoDB.	https://docs.aws.amazon.com/rds/ https://docs.aws.amazon.com/dynamodb/
Lab Practical (Syllabus covered from 46 to 90)		

10. Action plan for different types of learners

Slow Learners	Average Learners	Fast Learners
• Remedial Classes • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Expert Lecture • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design solutions for complex problems • Presentation on topics beyond those covered in CHO

11. Evaluation Scheme & Components

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
Component 2	Sessional Tests (STs)	02*	40%	Online/On Campus
Component 3	End Term Examination	01	60%	Online/On Campus
Total		**100%**		

* Out of 02 STs, the score of Best 1 ST will be considered.

* Practical File submission is mandatory.

* Makeup Examination will compensate for either ST-1 or ST-2 (Only for genuine cases, based on the Dean's approval and the exam will be based on the full syllabus).

**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

Syllabus of the Course:

S. No.	Contents	No.of sessions	Weightage (%)
1	Cloud Architecture: Layers and components of cloud services, Deployment Models: Public, Private, Hybrid, Community with comparison, Service Models: IaaS, PaaS, SaaS and their differences, Cloud Data Lifecycle.	4	5%
2	Virtualization: Introduction to virtualization techniques, Characteristics of virtualization, Pros and Cons of virtualization technology, Hypervisors and Types of Hypervisors, Multitenancy, Elasticity and Scalability, VirtualBox.	5	6%
3	AWS Learner Lab Environment & Console: Overview of AWS Academy Learner Lab, Lab session duration, budgets and limitations, Starting, stopping and resetting lab sessions, AWS Management Console overview, Regions and Availability Zones.	4	7%
4	Identity and Access Management (IAM) : Introduction to IAM concepts, Users, groups, roles and policies (conceptual overview), Preconfigured IAM roles in Learner Lab, Root account restrictions, Shared Responsibility Model.	7	6%
5	Compute Services: Introduction to Amazon EC2, Instance types, Launching and managing EC2 instances, Instance lifecycle management, Amazon Machine Images (AMIs), Auto Scaling, Elastic IP, Elastic Load Balancer and its Types, AWS CloudShell.	6	10%
6	Storage Services : Amazon S3 overview, Buckets and objects, S3 storage classes, Basic S3 permissions, Amazon EBS volumes, Attaching and managing EBS with EC2, Amazon EFS, Creating and mounting EFS with EC2, Amazon Glacier Storage.	7	10%
7	Containerization: Introduction to containerization, Docker concepts and architecture, Containers vs Virtual Machines, Kubernetes, Cloud security principles, Data protection strategies, Backup and recovery concepts, Cost awareness in AWS.	7	10%

8	Networking Services: Introduction to AWS Networking, Virtual Private Cloud (VPC), Subnets and CIDR blocks, Route Tables and Internet Gateway, Security Groups, Network Access Control Lists (NACLs), Route 53 (DNS System), AWS Direct Connect, Transit Gateway, CloudFront.	5	8%
9	AWS Security & Application Deployment: AWS Security: Introduction to AWS Security, AWS Key Management Service (KMS), Securing data at rest and in motion, Compliance issues, Privacy and security.	8	6%
10	Application Deployment: AWS Elastic Beanstalk overview, Application deployment workflow, Environment creation and management, Application versioning basics.	5	6%
11	Messaging and Integration Services: Amazon Simple Notification Service (SNS), Amazon Simple Queue Service (SQS), Event-driven architecture concepts, Message-based communication, AWS CloudTrail, AWS CloudWatch and Alarms.	5	5%
12	Introduction to DevOps in Cloud: Definition and principles of DevOps, Role of DevOps in cloud environments, Key DevOps practices: CI/CD, Automation, Monitoring, Collaboration.	6	8%
13	Database Services: Overview of AWS managed databases, Amazon RDS (MySQL / PostgreSQL), Database creation and connectivity, Amazon DynamoDB fundamentals, Data persistence across lab sessions, Amazon Redshift, Amazon Aurora	10	4%
14	Data Analytics Tools in AWS: Amazon Athena, AWS Glue, Amazon QuickSight, Amazon EMR, AWS Data Pipeline, Lake Formation, Amazon Kinesis, Amazon Timestream, Amazon SageMaker.	6	4%
15	AWS Developer & Migration Tools AWS X-Ray, AWS Cloud9, AWS CodeBuild, AWS CodeCommit, AWS CodePipeline, Introduction to Migration Tools, AWS Database Migration Service (DMS), AWS Server Migration Service (SMS), AWS Migration Hub.	5	5%
End Term Examination (ETE-Complete Syllabus)			

This document is approved by:

Designation	**Name**	**Signature**
Course Coordinator	Dr. Priyanka Gupta	
Head Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date (DD/MM/YYYY)	30/01/2026	